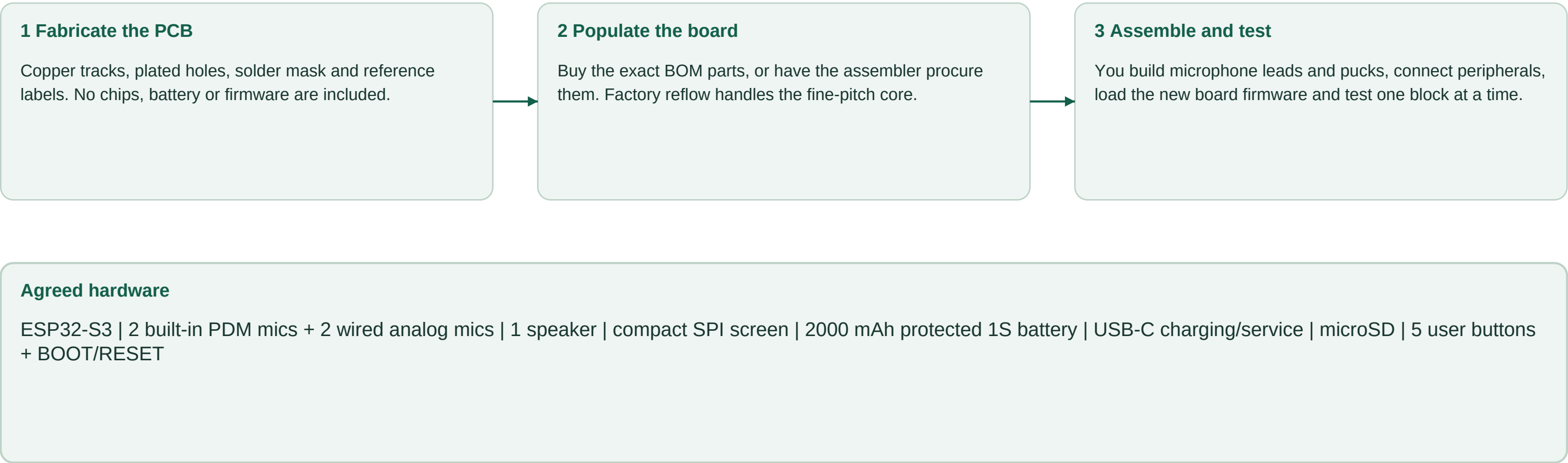


Your first custom Quipus board

A readable circuit draft, complete component inventory and practical route from an empty PCB to a tested recorder.



What this package supplies

Editable KiCad pin/net schematics, printable diagrams, main-board BOM, grouped purchase quantities, off-board microphone/tool lists and the assembly/change guides.

What still separates this draft from a board order

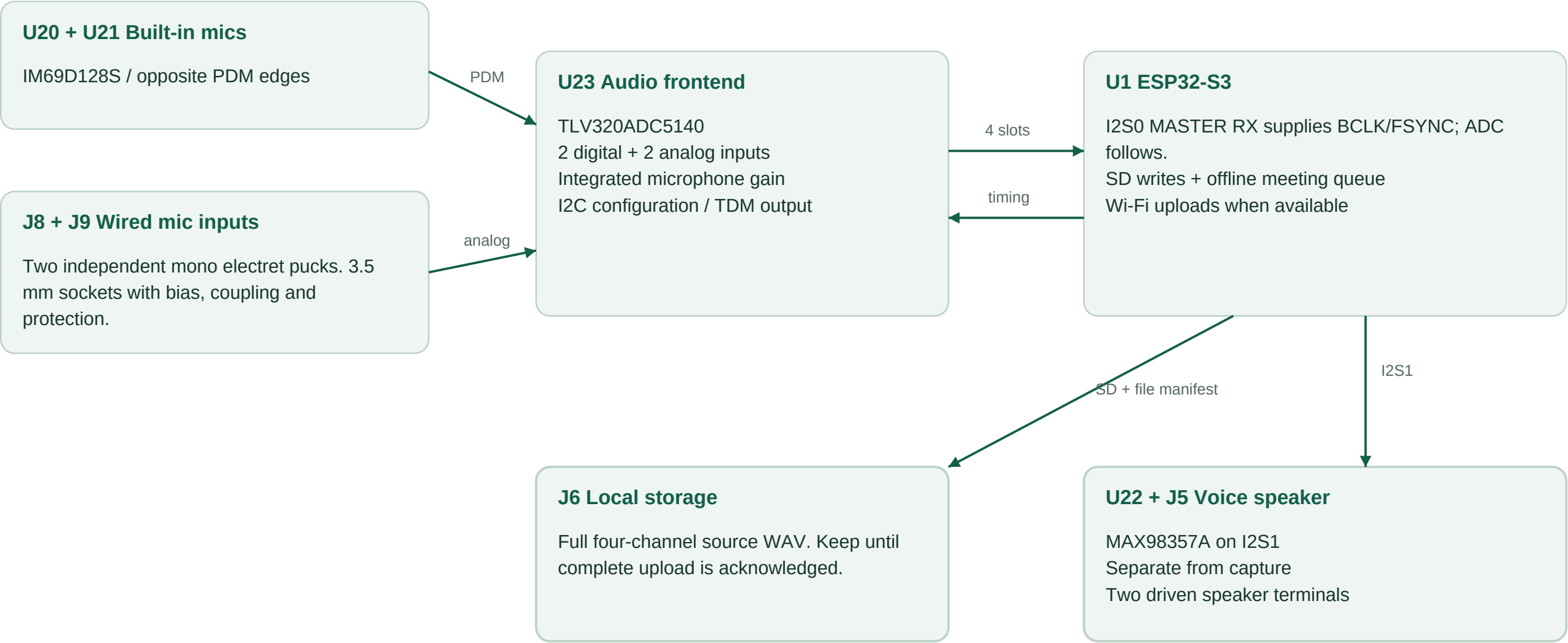
Exact footprints, battery/NTC approval, final passive part numbers, PCB routing, native ERC/DRC, enclosure fit and first-article electrical measurements. No fabrication files are included.

Best first build

Use selective factory assembly for the main SMD electronics. Learn soldering on practice boards and the two external microphone pucks, then assemble and diagnose your own recorder.

Four microphones, one synchronized capture path

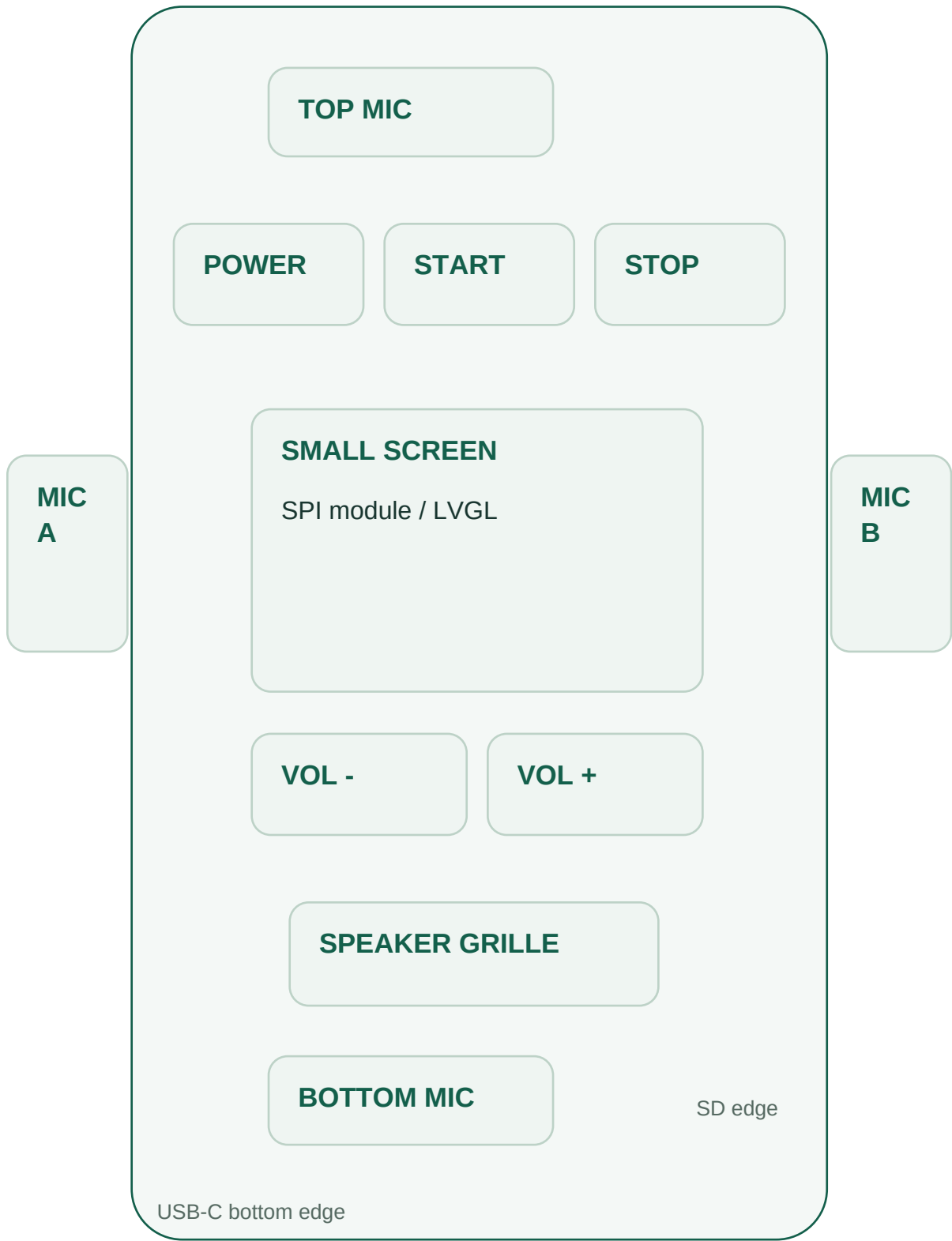
Near microphones stay on the device. Wired pucks extend placement; each source keeps its own channel.



The ADC is a hardware choice, not a speaker-identification algorithm. Preserve the source channels for later alignment, diarization and transcription. Two mic pucks are not combined with a splitter. USB-C is charging and USB device/service, not a general USB microphone host in Rev B.

Changes to the PCB and enclosure

Placement is conceptual. No pad coordinates, final outline or routing can be inferred from this drawing.



Audio change

Add U23 ADC and U24 PDM clock buffer. Replace the old short-wire PDM accessory header with J8/J9 analog mic sockets. Keep the two built-in mics and speaker amp.

Controls

Five everyday controls: Power, Start/Enter, Stop/Back, Volume up/down. Two recessed service controls: BOOT and RESET. Current B3U switches are SMD; alternative hand-solder buttons require new footprints.

Acoustics and layout

Bottom-port MEMS sensors need aligned unplated PCB holes and a short front acoustic passage. Separate the speaker chamber, analog inputs and switching-power loops. Preserve the module antenna keepout.

Mechanics

One mic socket on each side, USB-C accessible below, and clear microSD ejection space. Battery dimensions, jack bodies, plugs, cable bends and button travel must fit a new 3D assembly.

Neat engineering structure

Use functional schematic sheets, labeled nets and connector pin numbers. Use a reviewed four-layer stackup with a continuous ground reference, test pads and clear polarity/channel silkscreen.

Core parts to populate on the main PCB

These are selected candidates. The CSV includes every resistor, capacitor, switch and connector, with purchase holds.

Reference	Quantity	Manufacturer part candidate	Purpose / assembly
U1	1	ESP32-S3-WROOM-1-N16R8	Processor, Wi-Fi, 16 MB flash / 8 MB PSRAM
U20 / U21	2	IM69D128SV01XTMA1	Built-in digital MEMS microphones; factory reflow
U23	1	TLV320ADC5140IRTWR	Four-source audio frontend; WQFN / exposed pad
U24	1	SN74LVC1G17DBVR	PDM clock edge buffer; timing needs bench validation
U22	1	MAX98357AETE+	Digital speaker amplifier; speaker itself is off-board
U2 / U3	1 each	BQ24074RGTR / TUSB320LAIRWBR	USB power-path charging and Type-C current entitlement
U9 / U10	1 each	LTC2951ITS8-1#TRPBF / TPS22918DBVR	Power-button controller and main load switch
U11 / L1	1 each	TPS63802DLAR / XFL4015-471ME	3.3 V buck-boost regulator and 0.47 uH inductor
U12 / R19	1 each	BQ27441DRZR-G1A / 10 mOhm shunt	Battery measurement; actual cell profile must be configured
U30 / U31	1 each	RV-3028-C7-TA-QC / PCA9536DR	Timekeeping and control input expander
U4-U8	5	TLV75533PDBVR + LVC logic (see BOM)	Always-on supply and safe charging control logic
U13 / U34 / U35 / D1	4	USB protection / TS3USB221A / ESD	USB protection and off-state isolation; see exact BOM variants

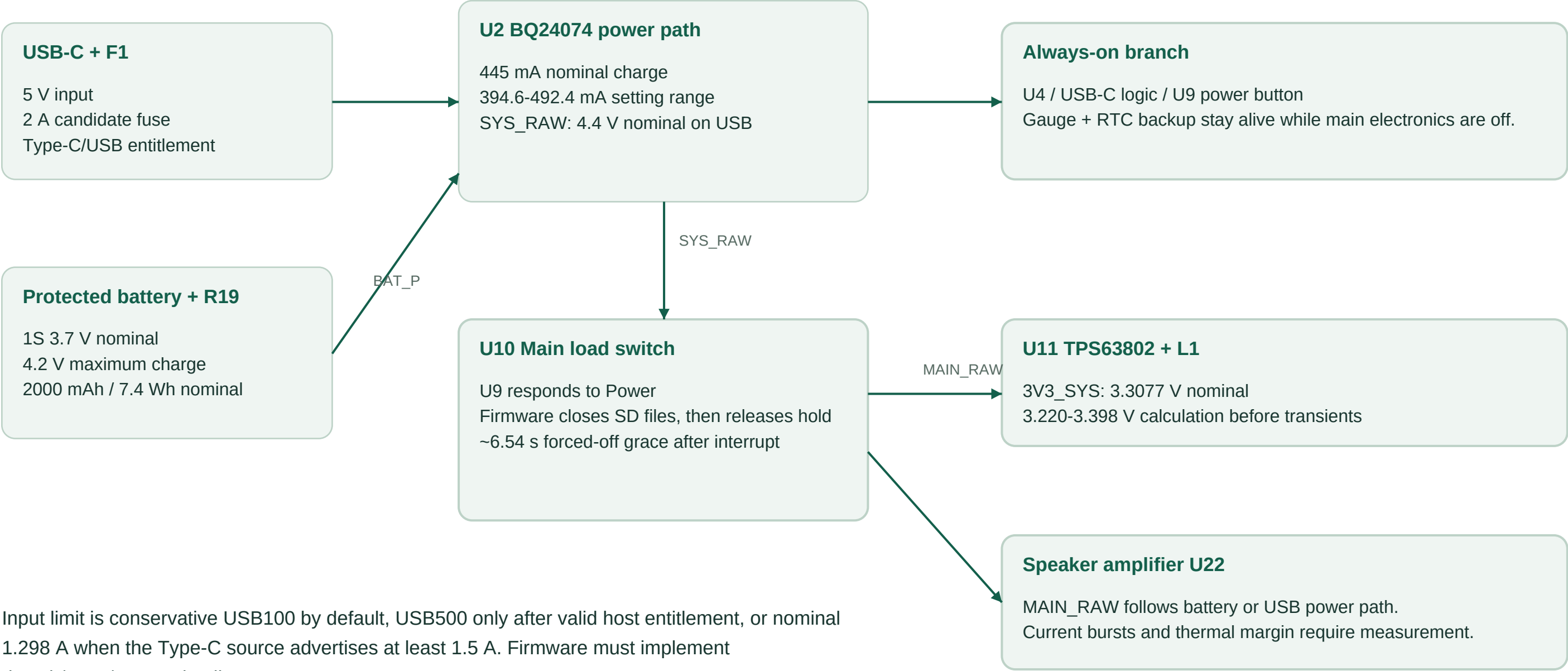
Connectors, controls and parts outside the PCB

Do not buy by appearance. The part package and connector contact order determine whether it fits.

Reference	Qty	Part / requirement	Fit and purchase notes
J1	1	GCT USB4105-GF-A	USB-C charge + native USB; factory assembly
J2	1	JST S2B-PH-SM4-TB(LF)(SN)	Battery PH 2 mm / SMD; check pin 1 positive
J3	1	3-position JST SH NTC connector - HOLD	Exact MPN/footprint and battery temperature window pending
J5	1	SM02B-SRSS-TB	Speaker connector; SMD, not a breadboard part
J6	1	Hirose DM3AT-SF-PEJM5	Push-push microSD socket; factory soldering
J7	1	JST B8B-PH-K-S(LF)(SN)	8-position display header; through-hole hand option
J8 / J9	2	Same Sky SJ1-3533NG	3.5 mm TRS through-hole mic sockets; exact pad map specified
SW1	1	Momentary power switch - HOLD	Exact switch and footprint need release
SW300-SW305	6	Omron B3U-1000P	SMD: BOOT, Start, Stop, Up, Down, RESET
Off-board	2	PUI AOM-5024L-HD-R	External electret capsules, shielded leads and 3.5 mm plugs
Off-board	1 each	Protected 1S 2000 mAh LiPo + cell-mounted NTC	Exact pack, dimensions and charging window are HOLD
Off-board	1 each	Waveshare 1.3-inch display / 8 ohm 2 W speaker / SD	Verify display revision; exact speaker HOLD; genuine test card

Voltage domains and charging current

Raw battery and USB power never connect directly to 3.3 V GPIO, microphone, SD or display rails.



Input limit is conservative USB100 by default, USB500 only after valid host entitlement, or nominal 1.298 A when the Type-C source advertises at least 1.5 A. Firmware must implement detach/reset/suspend policy.

Open approval: the proposed charger/thermistor combination has a nominal 0-50 deg C window. A battery rated for 0-45 deg C charging needs a tighter hardware window and tolerance review. Charger dissipation can be about 0.8 W at 5 V -> 3.2 V / 445 mA, before other heat.

Connector contact definitions

Physical contacts from the exact selected part. Matching net names connect across all sheets; NC is left unconnected.

J1 / GCT USB4105-GF-A

A1_B12: GND -> GND | A4_B9: VBUS -> VBUS_5V | B8: SBU2 -> NC | A5: CC1 -> USB_CC1 | B7: D-2 -> USB_DM | A6: D+1 -> USB_DP | A7: D-1 -> USB_DM | B6: D+2 -> USB_DP | A8: SBU1 -> NC | B5: CC2 -> USB_CC2 | B4_A9: VBUS -> VBUS_5V | B1_A12: GND -> GND | SH1: SHIELD -> GND | SH2: SHIELD -> GND | SH3: SHIELD -> GND | SH4: SHIELD -> GND

Sixteen mating contacts map to twelve solder tails: A1/B12, A4/B9, B4/A9, B1/A12 share pads. Four shield stakes connect to ground. Pad names below explicitly combine shared contacts; verify the imported CAD footprint uses the same mapping. No external CC Rd resistors.

J2 / JST S2B-PH-SM4-TB(LF)(SN) battery

1: PACK+ -> BAT_PACK_P | 2: PACK- -> GND

Board-defined pin 1 is positive. Verify pack polarity physically. Protected 1S 2000 mAh pack, standard 4.2 V charge, at least 2 A discharge capability. Swap with device off and USB unplugged.

J3 / 3-pin keyed NTC connector - JST SH family candidate

1: NTC -> BAT_NTC | 2: NTC_RETURN -> GND | 3: KEY_NC -> NC

Three positions prevent swapping with the two-position speaker connector. A 10 kOhm 103AT-2 thermistor must be thermally attached to the cell. Do not populate a fixed bypass resistor.

Connector contact definitions

Physical contacts from the exact selected part. Matching net names connect across all sheets; NC is left unconnected.

J6 / DM3AT-SF-PEJM5

1: DAT2 -> SD_D2_PU | 2: CD_DAT3 -> SD_D3_PU | 3: CMD -> SD_CMD | 4: VDD -> 3V3_SYS | 5: CLK -> SD_CLK | 6: VSS -> GND | 7: DAT0 -> SD_D0 | 8: DAT1 -> SD_D1_PU | A: CARD_DETECT_A -> SD_CD_N | B: CARD_DETECT_B -> GND | SH: SHIELD -> GND

Native 1-bit SDMMC. DAT1/2/3 have 10k pull-ups even though not routed to ESP. Hirose drawing names separate detect contacts A/B, not 9/10; CAD footprint must map those named pads explicitly. Symbol SH represents all shell solder stakes; do not short detect-A to shell through a generic footprint. Detect closes A-B with card inserted. Enclosure must allow ejection travel.

J7 / B8B-PH-K-S(LF)(SN)

1: BL -> LCD_BL | 2: RST -> LCD_RST_N | 3: DC -> LCD_DC | 4: CS -> LCD_CS_N | 5: CLK -> LCD_SCLK | 6: DIN -> LCD_MOSI | 7: GND -> GND | 8: VCC -> 3V3_SYS

Custom Quipus connector numbering matches Waveshare SKU15867 MODULE J2 PH2.0, not its reversed J1 header. Vendor schematic J2:1BL,2RST,3DC,4CS,5SCK,6DIN,7GND,8VCC. Match signals by continuity, do not assume supplied cable colors/orientation. External module 45x31mm includes ST7789 240x240 and NPN backlight driver; no direct LED-current drive from GPIO.

J5 / SM02B-SRSS-TB

1: SPK+ -> SPK_P | 2: SPK- -> SPK_N

Mates SHR-02V-S. Neither pin is ground. Short twisted pair to 8 ohm, 2 W nominal dynamic speaker; exact speaker part pending. Speaker connector intentionally differs from battery series.

Connector contact definitions

Physical contacts from the exact selected part. Matching net names connect across all sheets; NC is left unconnected.

J8 / SJ1-3533NG

1: SLEEVE -> GND | 2: TIP -> MIC1_JACK_TIP | 3: RING -> NC

Manufacturer sheet2 pinmap verified:1 sleeve,2 tip,3 ring. Ring intentionally NC for own mono accessory. No insertion-detect switch. Create/verify exact footprint and1:1 paper fit before fabrication. This is MIC IN, not headphone output.

J9 / SJ1-3533NG

1: SLEEVE -> GND | 2: TIP -> MIC2_JACK_TIP | 3: RING -> NC

Manufacturer sheet2 pinmap verified:1 sleeve,2 tip,3 ring. Ring intentionally NC for own mono accessory. No insertion-detect switch. Create/verify exact footprint and1:1 paper fit before fabrication. This is MIC IN, not headphone output.

Audio supply, gain and timing review

Nominal values and planning allowances are distinguished from absolute limits and measured behavior.

Audio node / function	Voltage or allocation	Engineering requirement
U23 AVDD / pin 1	Filtered 3.3 V; permitted 3.0-3.6 V	FB220 supply bead + local bypass. Its 220 ohm high-frequency impedance is not its DC resistance.
U23 IOVDD / pin 19	3.3 V; permitted 3.0-3.6 V	Matches ESP32 digital levels. Bypass close to the ADC pin.
AREG pin 2 / DREG pin 24	Internal 1.8 V / 1.5 V outputs	Their dedicated capacitors return to GND. Never tie these regulator outputs to 3.3 V.
VREF pin 3 / MICBIAS pin 5	~2.75 V reference / configured 3.014 V bias	Reference settling and effective capacitor value need verification. MICBIAS published limit is 20 mA.
U20 / U21 / U24 supply	MIC_3V3, nominal 3.3 V	Two MEMS sensors plus buffer; local bypass; load/edge measurements required.
Capture subsystem allowance	37 mA x 3.3 V = 122.1 mW	Planning: ADC 30 mA + two mics 2 mA + buffer 2 mA + bias allowance 3 mA. Not measured or a guaranteed maximum.
Gain and input range	Start at 0 dB / 20 kOhm; up to 42 dB PGA	ADC single-ended AC full scale is 1 Vrms at its specified conditions. Increase gain only after level/noise/clipping tests.

PDM timing is not closed yet

The sensor requires <=13 ns clock edges while the ADC allows up to 18 ns at its PDM output. U24 is an edge-restoring candidate; measure loaded duty cycle, edge, propagation and data setup/hold. Begin at 1.536 MHz if needed and qualify 3.072 MHz separately.

External-input protection is a release hold

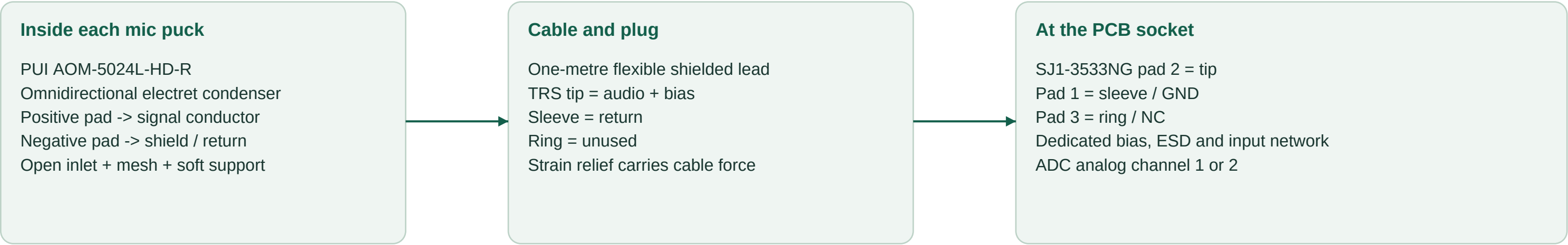
The candidate socket TVS pulse clamp exceeds the ADC absolute input maximum. The full series/coupling/protection network needs transient qualification or redesign before manufacturing release; a TVS symbol alone is not proof of protection.

Whole-device power still needs measurement

The inherited 0.85 A 3.3 V peak envelope was not requalified for Rev B. New audio load, Wi-Fi, SD, display and speaker bursts must be measured together, including battery-path drop, switch margin and enclosure heating.

Wired microphone puck: how it is connected

One capsule per socket. This is our defined mono plug-in-power accessory, not universal compatibility with every headset.



Bias and gain

The board supplies approximately 3.014 V through each 2.2 kOhm bias resistor. The capsule has about 0.5 mA datasheet consumption at its stated test condition; the resistor drop is about 1.1 V. Exact operating current and signal level must be measured. The ADC provides input gain, so a separate puck preamplifier is not proposed for the first one-metre test.

Clear audio is a system result

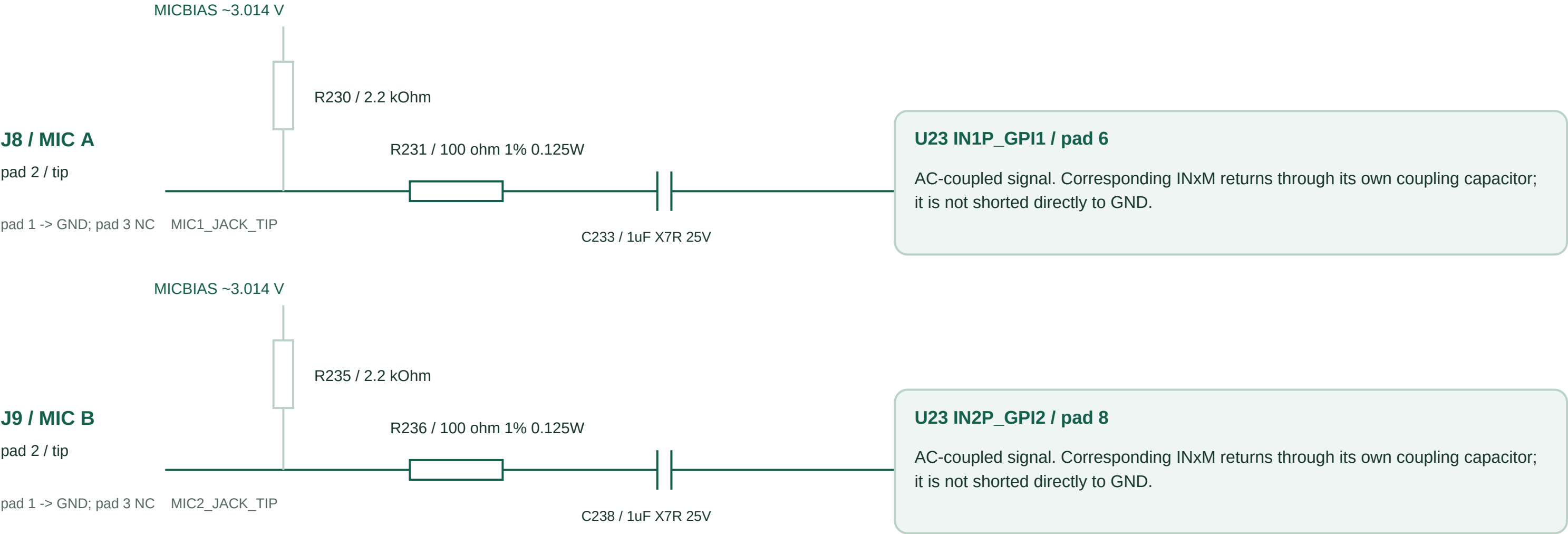
The capsule datasheet lists -24 +/-3 dBV/Pa sensitivity and 80 dB A-weighted SNR under its test conditions. These are not a guaranteed room pickup distance. Speech level, reverberation, placement, cable interference and ADC clipping still determine the recording.

What you solder

Practice first. The published capsule drawing does not label the physical positive/negative pads: obtain manufacturer-confirmed polarity before soldering or applying power. Then tin the leads, briefly solder terminals and plug, and fit heat shrink/strain relief. PUI specifies 360 +/-10 deg C, <=2 s per terminal and an iron below 90 W. Never block the acoustic inlet.

Analog input circuit and digital channel map

Functional input schematic. Exact resistor/capacitor designators, all chip pads and their net names follow in the circuit appendix.



Source	ADC input / path	ESP32 / timing	Storage contract
Top + bottom digital MEMS	ADC channels 3/4; PDM data pair and buffered clock	GPIO4 BCLK output; GPIO5 FSYNC output	Two separate built-in source channels
Analog MIC A + MIC B	Independent analog channels 1/2; bias + AC input	GPIO6 TDM input; GPIO42 ADC shutdown	Two separate external source channels
Timing at 16 kHz	ADC slave PLL referenced to BCLK	4 slots x 32 bits x 16000 = 2.048 MHz	Firmware must verify actual slot order before WAV mapping

Order and assembly: the whole process

Do these stages in order. First learn on inexpensive parts; an empty custom PCB is not a soldering practice kit.

1 Lock the design

Approve the exact battery/NTC, speaker, power switch, connectors and passive MPNs. Review the schematic in CAD, assign exact footprints, finish placement/routing, and pass ERC/DRC plus mechanical/thermal review.

2 Get two assembly quotes

Send reviewed Gerbers/drill files, stackup, BOM, CPL/pick-and-place and assembly drawings. Ask for two populated prototypes and a spare bare PCB, selective assembly scope, substitutions, inspection and extra parts for attrition. PCBWay or a Malaysian assembly shop can quote this; a bare-board order alone supplies no components.

3 Obtain off-board parts and tools

The main-board shopping CSV is separate from battery, screen, speaker, SD and the two microphone accessories. Buy a temperature-controlled iron, solder/flux, stand, magnifier, multimeter, wire tools, heat shrink and a practice kit. Borrow a current-limited supply and scope for first-board checks.

4 Let the factory assemble the fine-pitch core

Factory-populate the ADC, MEMS mics, hidden-pad power chips, USB-C and SD socket, preferably all SMD parts. You can manually install the through-hole mic sockets/display header if explicitly left unassembled and after practice.

5 Build the pucks and connect harnesses

Solder one capsule/plug cable, verify continuity and shorts, test it, then make the second. Use precrimped JST harnesses and meter-check pin order before attaching screen, speaker or battery. Mount components without blocking acoustic ports.

6 Bring up and validate

Inspect unpowered, test power from a current-limited supply, then boot/USB, I2C, controls, display, SD, each microphone and speaker. Fit the approved battery last. Load a Rev B firmware profile and test complete WAV retention/upload before daily-use trials.

First power and first recording checks

Record measurements per serial-numbered board. A structural netlist check is not evidence of safe operation.

Stage	What to do	What to verify
Unpowered	Unplug battery/USB and inspect under magnification; meter-check polarity and stabilized rail resistance.	No solder bridges, blocked mic ports, swapped harnesses or hard shorts. Capacitors may cause a brief continuity beep.
Power only	Use a reviewed J2 harness and current-limited 3.7 V bench supply, no LiPo/USB; hold processor reset and remove peripherals.	BAT_PACK_P/BAT_P, SYS_RAW, 3V3_AON, MAIN_RAW when switched, and regulated 3.3 V. Scope ripple/overshoot.
Processor + USB + I2C	Use minimal custom-board diagnostics; verify BOOT/RESET, USB orientation/current policy and peripheral addresses.	RTC 0x52, gauge 0x55, expander 0x41, ADC 0x4C. USB current entitlement is implemented, not assumed.
Screen / controls / SD	Attach one verified load at a time while powered off; write/read a card test file before audio.	Five user controls plus two service controls. Quipus J7 to display-module J2 signal order; SD detection, write integrity and unmount.
Audio	Built-in mics separately, then MIC A, then MIC B, then all four; speaker at low level.	Actual channel identity, clipping/noise, PLL lock and PDM edges. Never ground a speaker output or scope clip.
Daily-use behavior	Capture/upload with Wi-Fi and playback variations; test Stop, reboot, queue recovery and loss of network.	Full source audio duration and all channels, local retention until confirmed upload, responsive buttons and measured battery/thermal behavior.

Battery charging begins only after exact cell/NTC approval and bench checks. A temporary 50 mA diagnostic limit with the MCU held in reset is not a usable operating current; do not raise it blindly to hide a fault.

Battery, file size and latency: numbers to test

Transparent calculations. No week-long runtime, far-field quality or upload-speed promise is implied.

Quantity	Calculation / value	Meaning
Nominal battery energy	2.0 Ah x 3.7 V = 7.4 Wh	Usable energy depends on load, cell condition, cutoff and losses.
Charging	2000 / 445 = 4.49 h ideal	Lower bound only; CV tail, system load, source limit and thermal regulation extend this.
Four-channel PCM16	16000 x 4 x 2 = 128000 bytes/s	7.68 MB/min; 460.8 MB/hour; 2.7648 GB for six hours.
I2S DMA before PCM16 conversion	16000 x 4 x 4 = 256000 bytes/s	32-bit transport containers; bounded DMA buffers, not whole meetings in RAM.
Single classic WAV/FAT32 file	~4 GiB / 128000 = 9.32 h	Segment before the file limit and preserve a meeting manifest. Filesystem choice alone does not remove RIFF limits.
35-42 h/week recording target	1600 mAh usable / 35-42 h = 46-38 mA average	A stringent battery-side total including everything; not supported by a measured Quipus result.

Assumed battery-side average	Runtime with 20% planning reserve	Recording days
80 mA	20.0 hours	3.33 six-hour days
120 mA	13.33 hours	2.22 six-hour days
180 mA	8.89 hours	1.48 six-hour days
240 mA	6.67 hours	1.11 six-hour days

A one-metre analog cable adds only an estimated few nanoseconds of electrical propagation, negligible for speech. Acoustic placement, ADC filter delay, DMA, SD and cloud processing matter more. Analog and PDM paths share a sample rate but require measured delay alignment before phase-sensitive mixing.

How to read the detailed schematic pages

A conventional signal flow is shown earlier. The following appendix gives every component and physical pad connection.

Reference labels

U = chip/module, R = resistor, C = capacitor, J = connector, L = inductor, D = protection device, SW = switch. Each reference appears exactly once in the editable schematic and BOM.

Named nets

Every matching global name is the same electrical connection, even on another sheet. For example I2C_SDA connects the processor, RTC, gauge, control expander and ADC. GND is a common reference plane. NC is intentionally unconnected.

Physical pins

Numbers identify the named manufacturer package contacts, not GPIO numbers or guessed community-library symbols. Follow the connector manufacturer drawing for its pad/contact mapping. Detailed pin-connections.csv is useful when checking a harness.

DNP and holds

DNP means do not populate during the normal first build. Hold rows need an exact qualified MPN/footprint or electrical/mechanical approval. A draft candidate is not authorization to substitute a similar-looking chip.

Validation actually performed

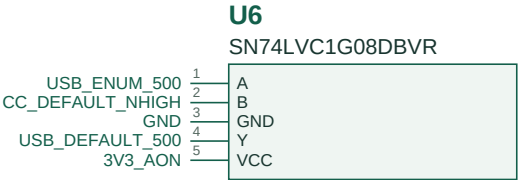
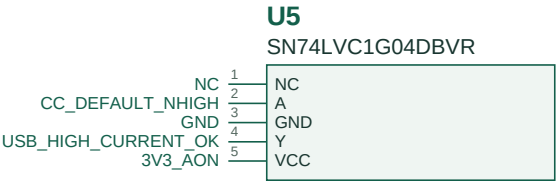
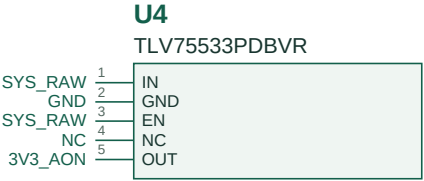
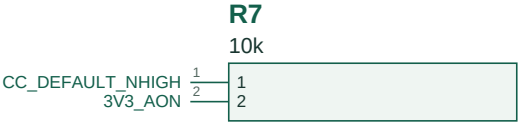
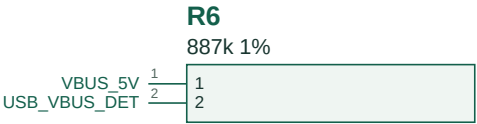
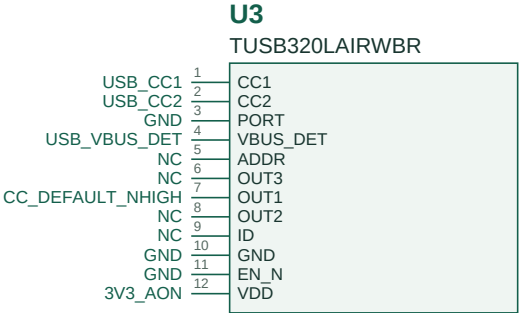
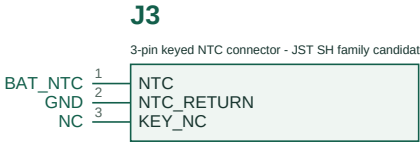
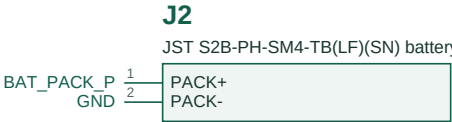
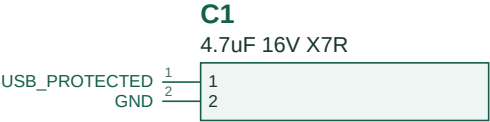
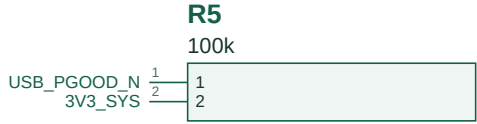
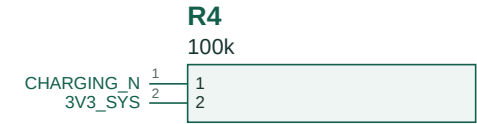
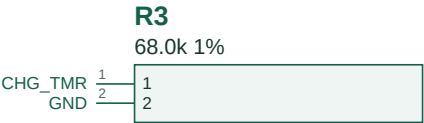
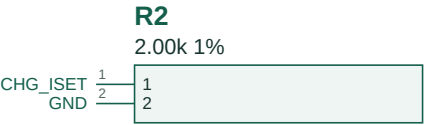
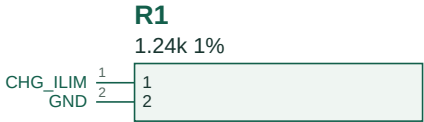
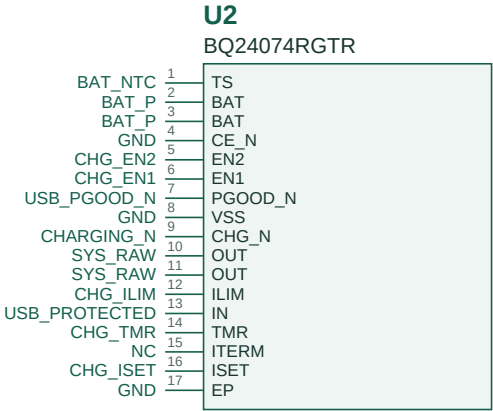
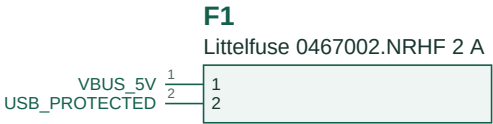
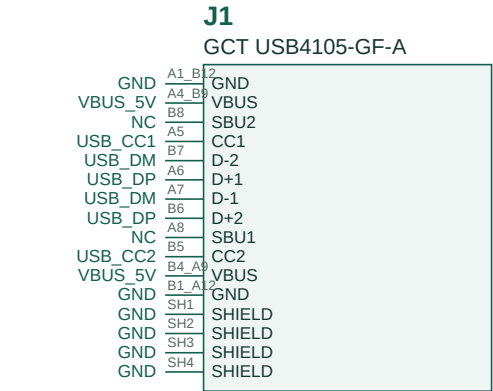
JSON parsing, unique references and physical pins, selected critical-net checks, source/export consistency and generated schematic syntax checks. Native KiCad/ERC and EasyEDA import have not been run here. Footprints, layout, DRC, electrical/thermal and acoustic tests remain required.

CAD files

Open schematics/Quipus-B1.kicad_sch with its child sheets. Symbols are embedded draft pin boxes and footprints are deliberately unassigned. Legacy sources are provided for a CAD engineer to attempt conversion. This package does not update the saved EasyEDA board.

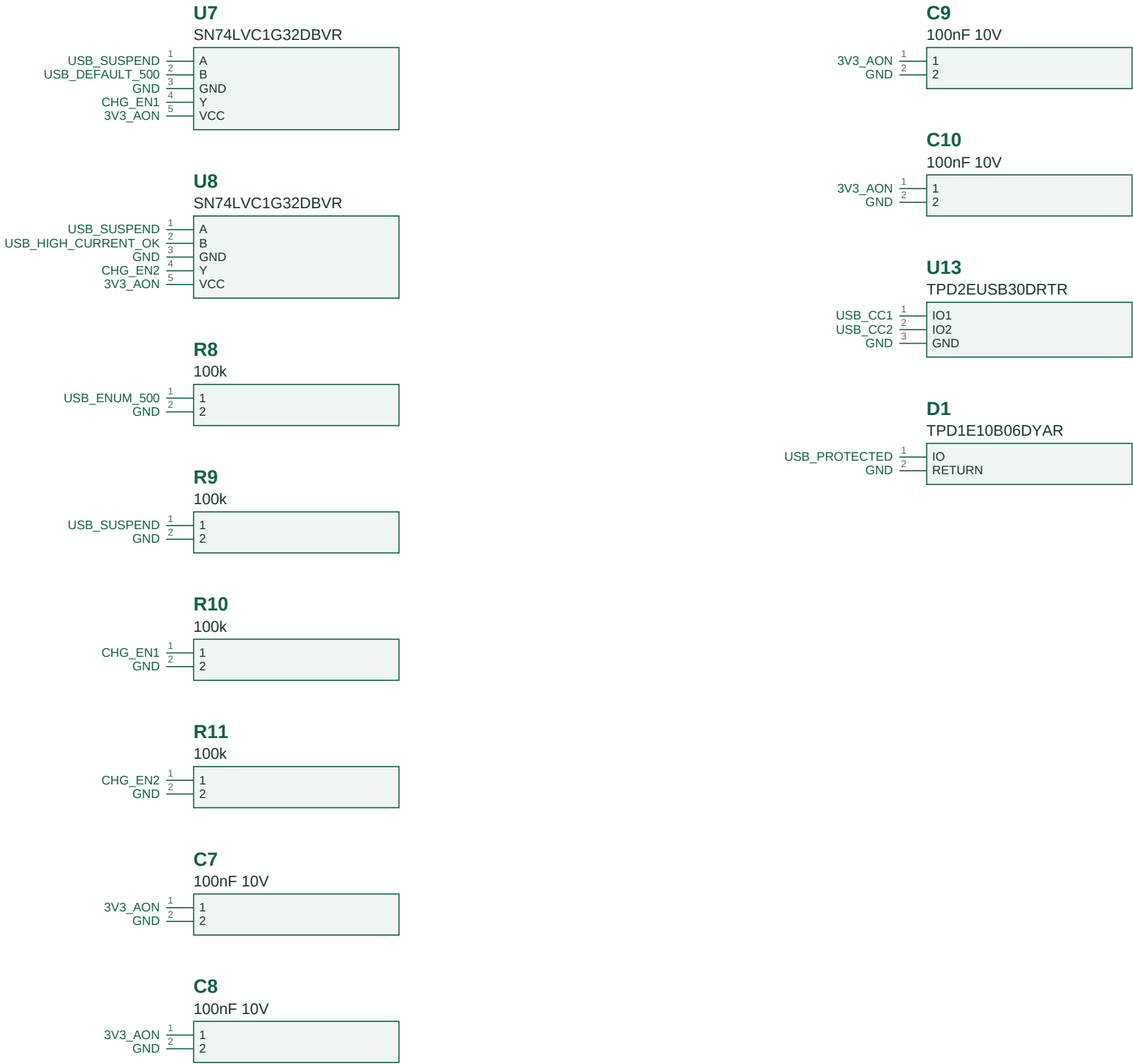
Circuit 01 / usb charge

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



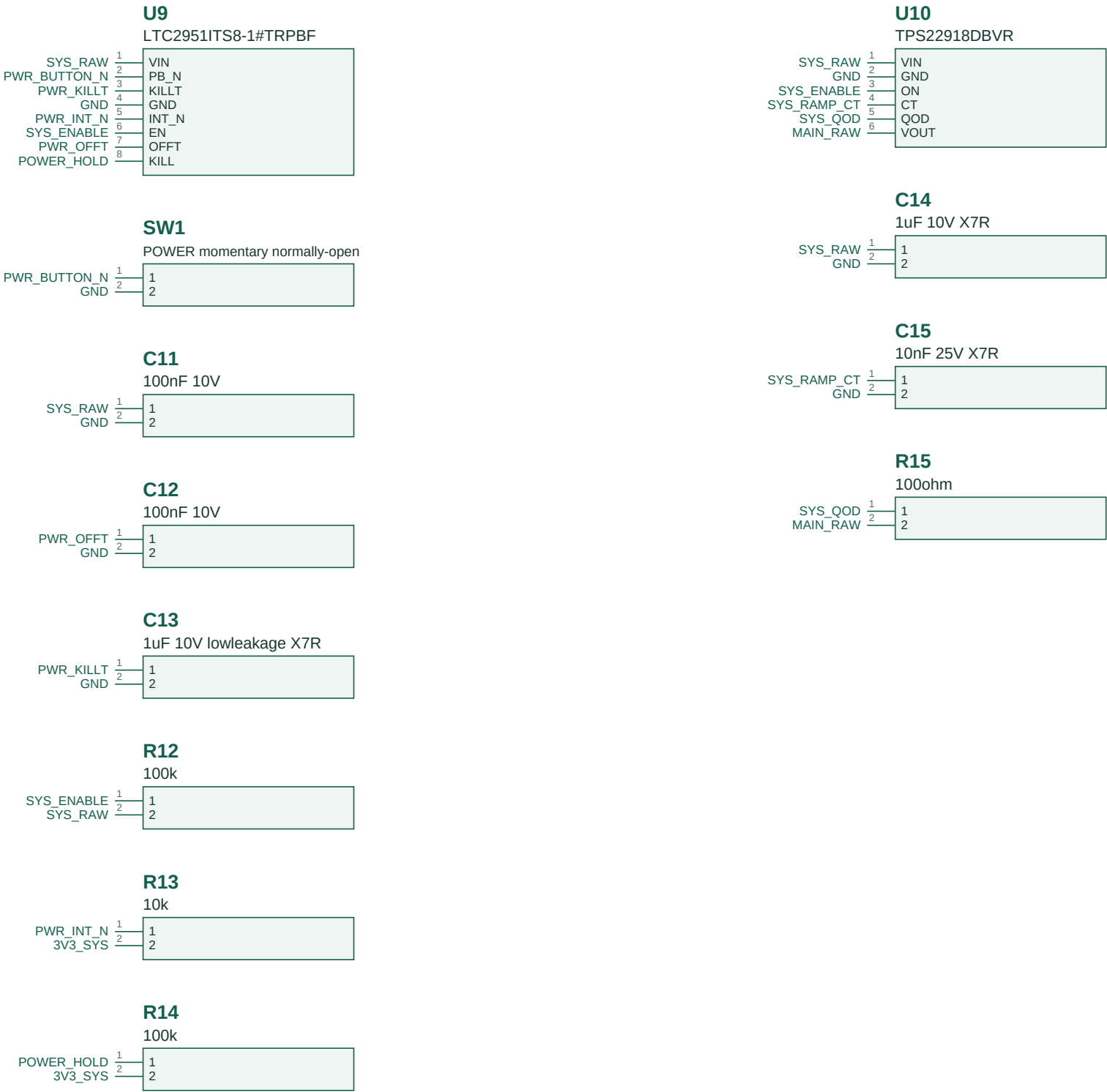
Circuit 02 / usb charge

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



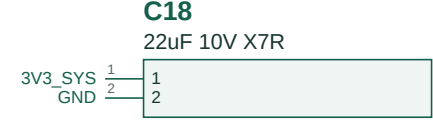
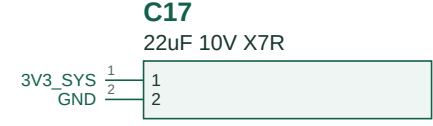
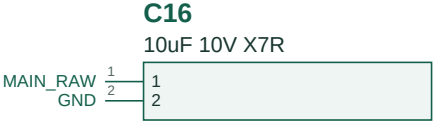
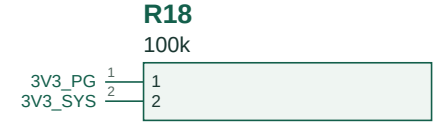
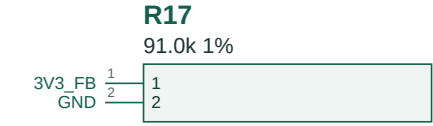
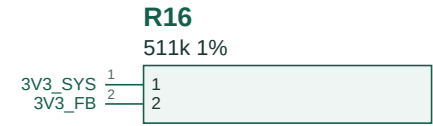
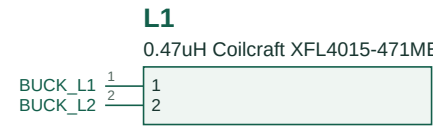
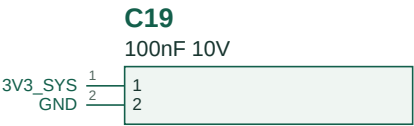
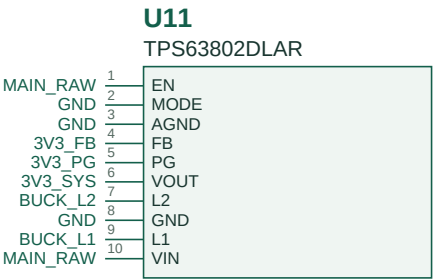
Circuit 03 / power control

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



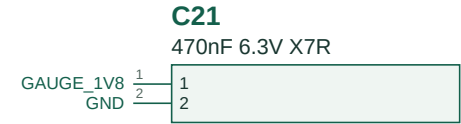
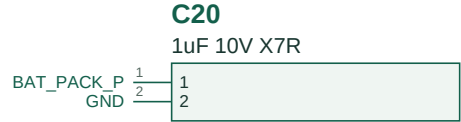
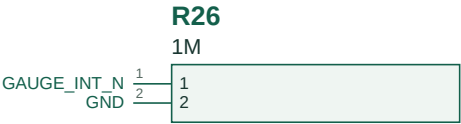
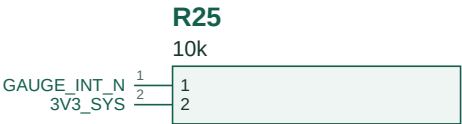
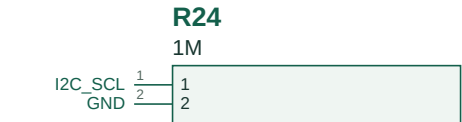
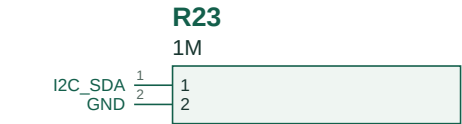
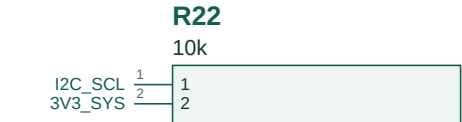
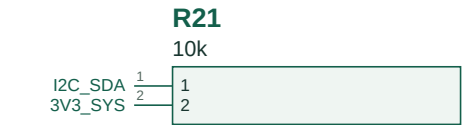
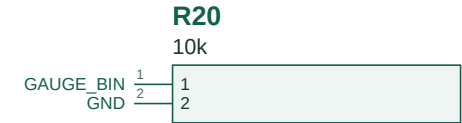
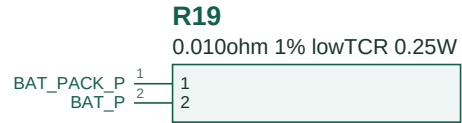
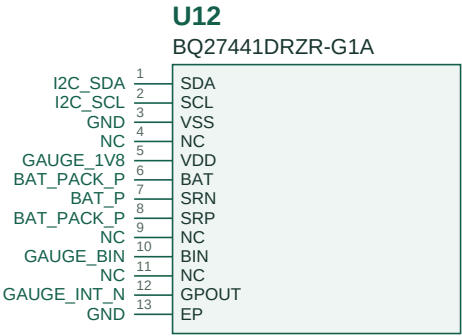
Circuit 04 / regulator

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



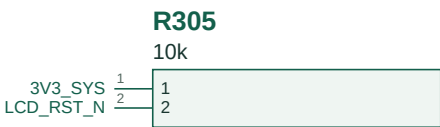
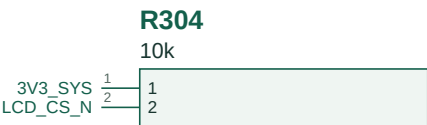
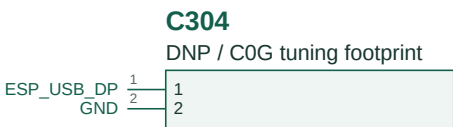
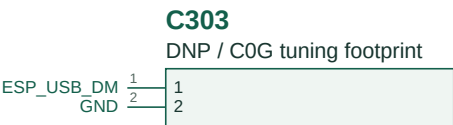
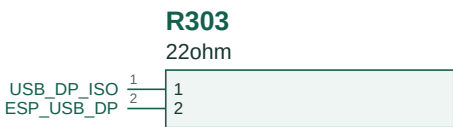
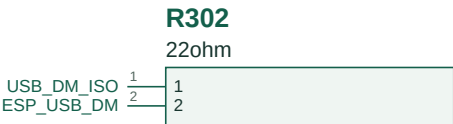
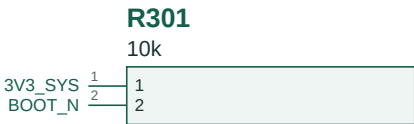
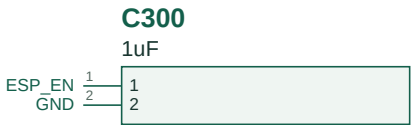
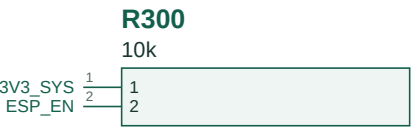
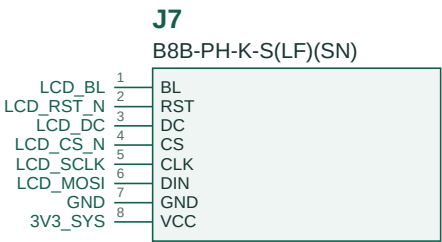
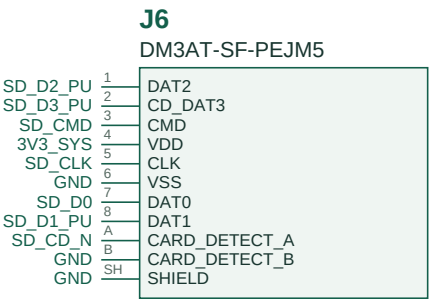
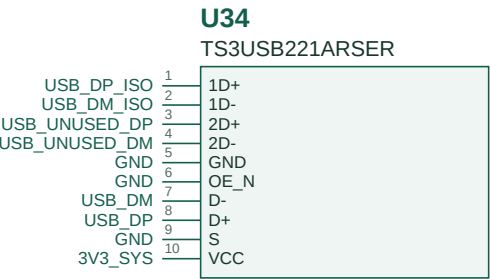
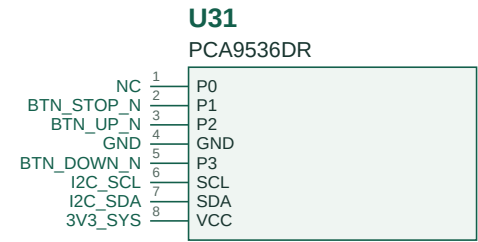
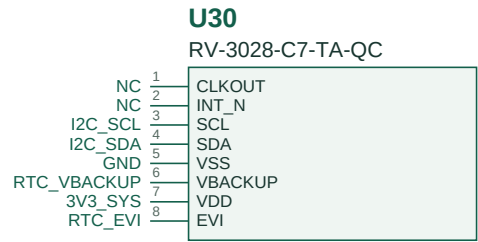
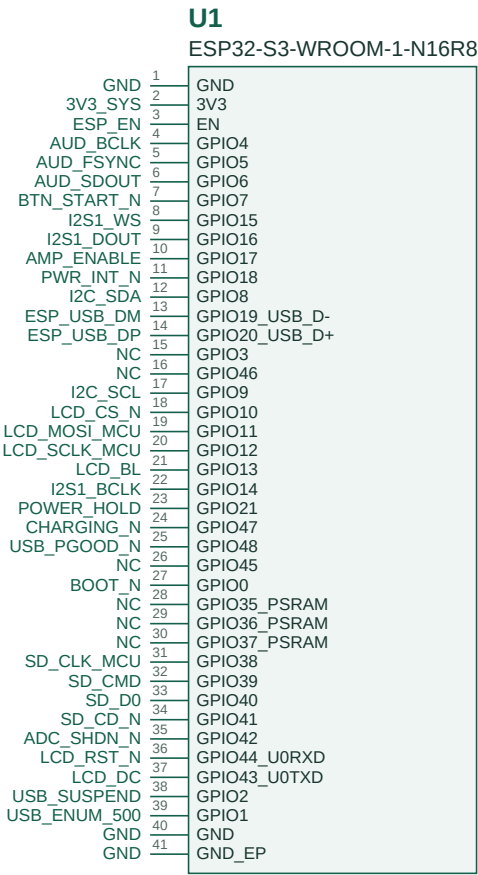
Circuit 05 / gauge

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



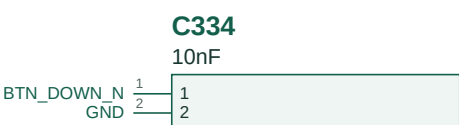
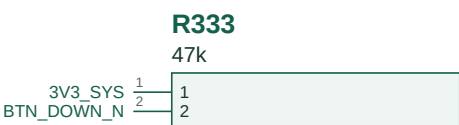
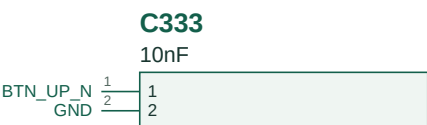
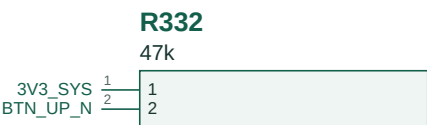
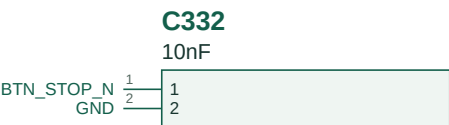
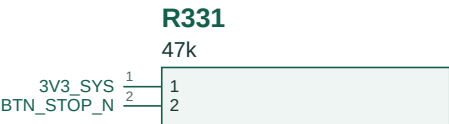
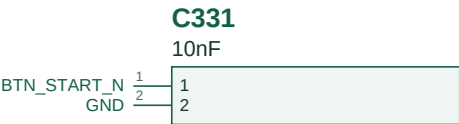
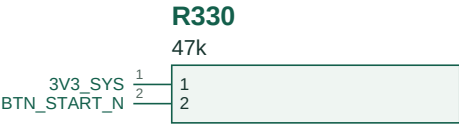
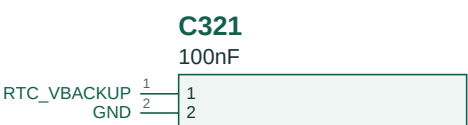
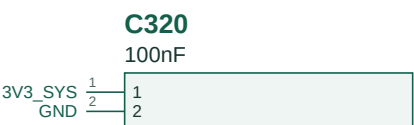
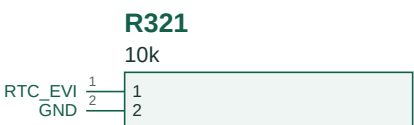
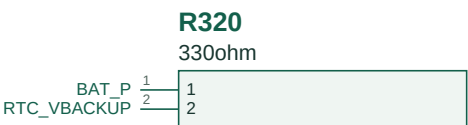
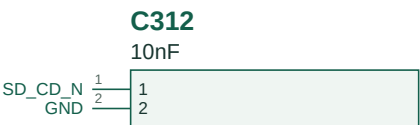
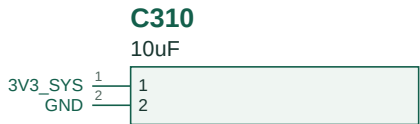
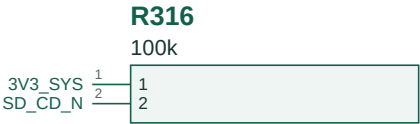
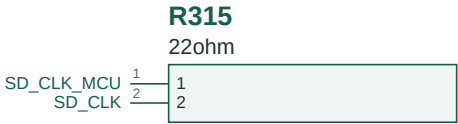
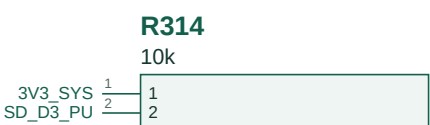
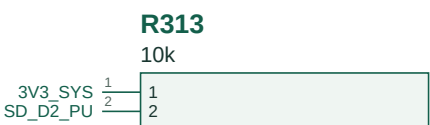
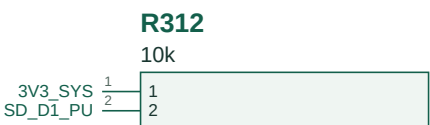
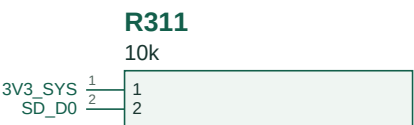
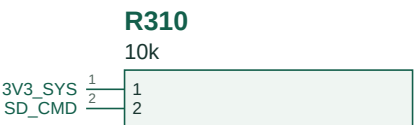
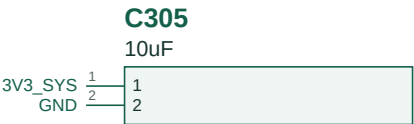
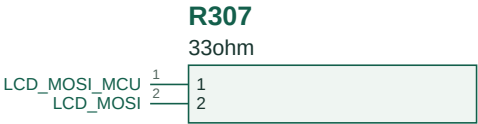
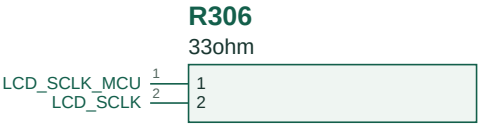
Circuit 06 / Controller

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



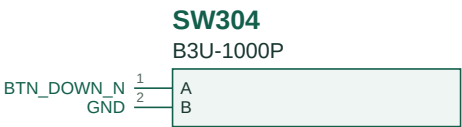
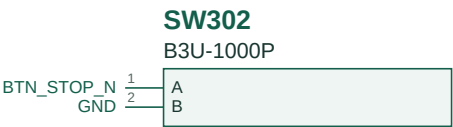
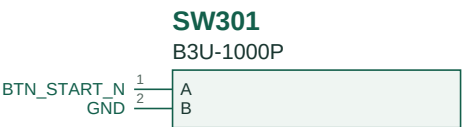
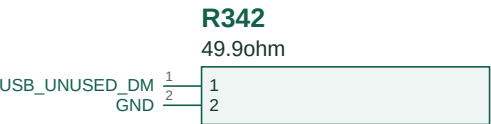
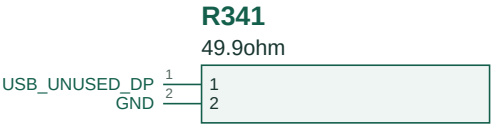
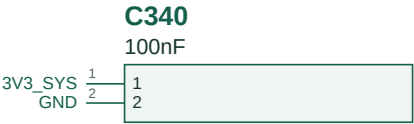
Circuit 07 / Controller

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



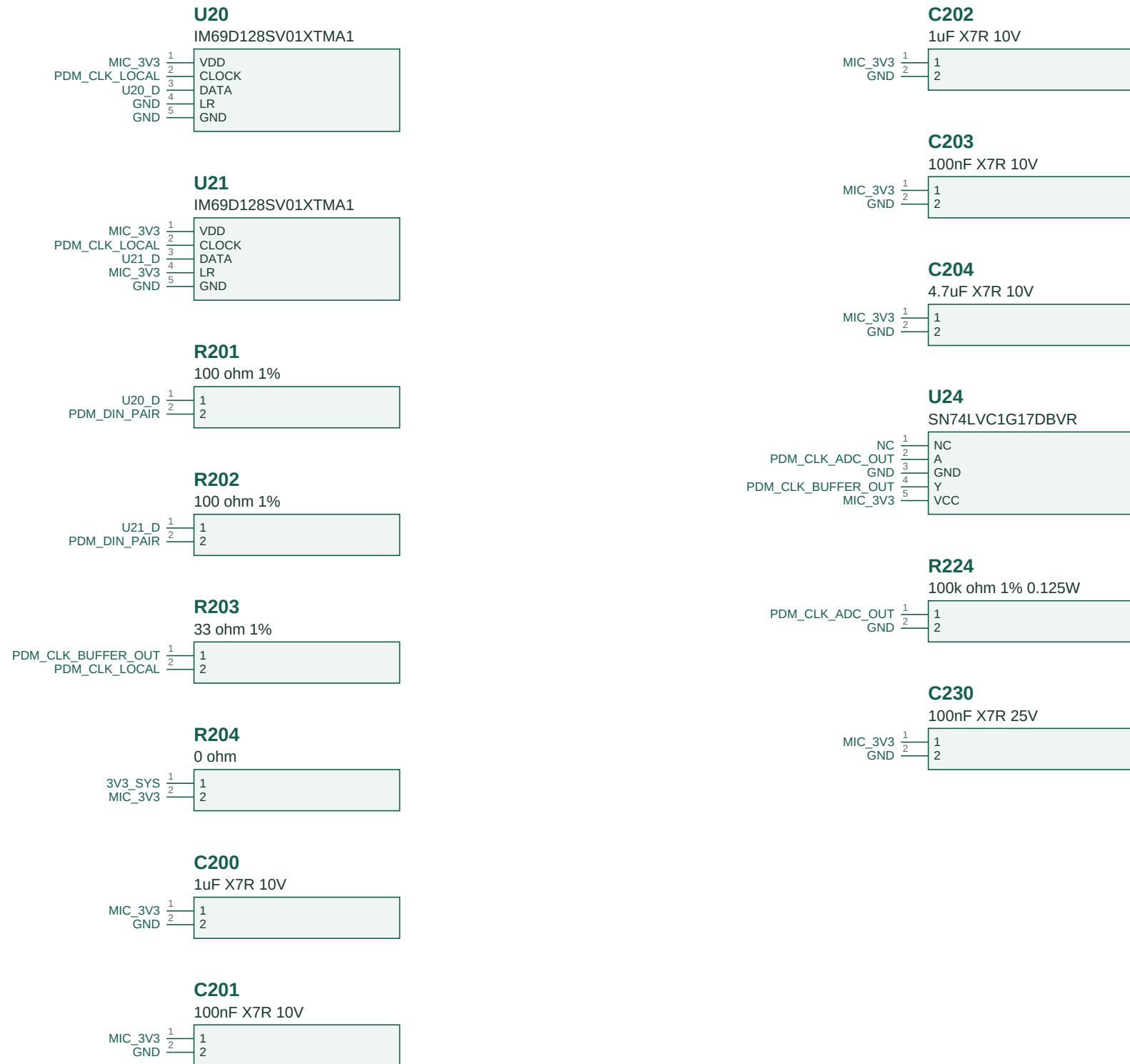
Circuit 08 / Controller

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



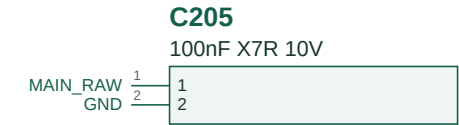
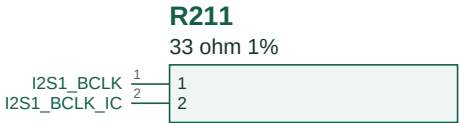
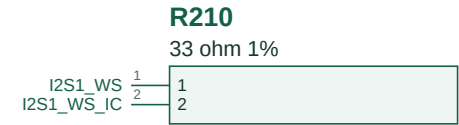
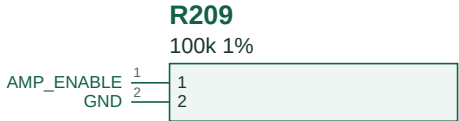
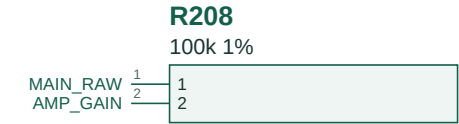
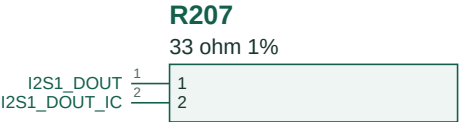
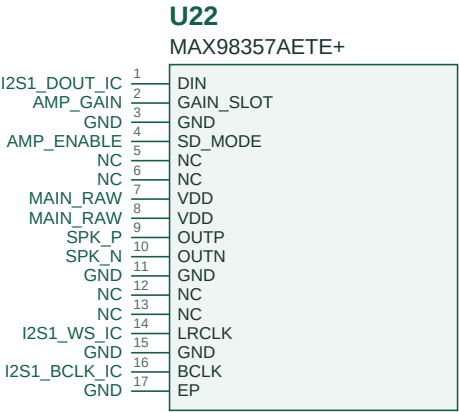
Circuit 09 / Internal microphones

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



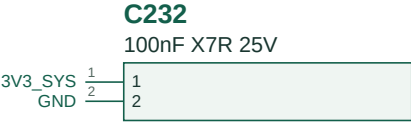
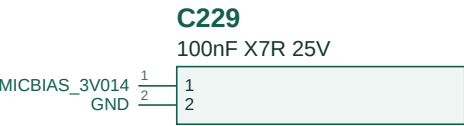
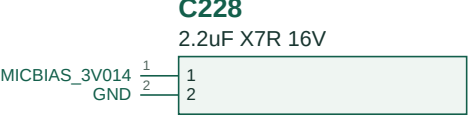
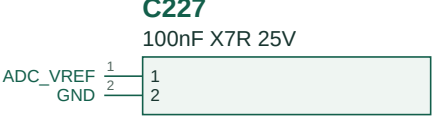
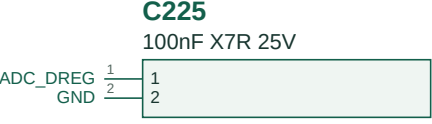
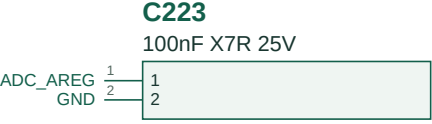
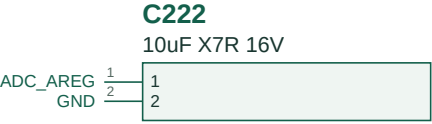
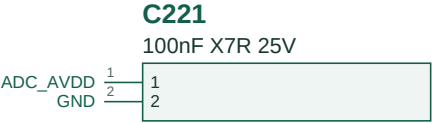
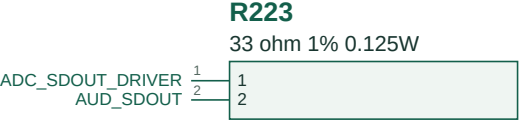
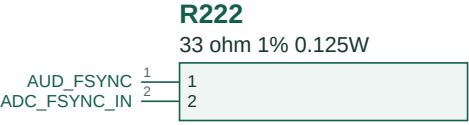
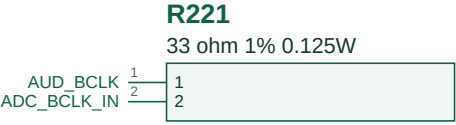
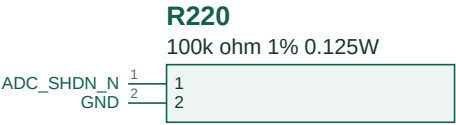
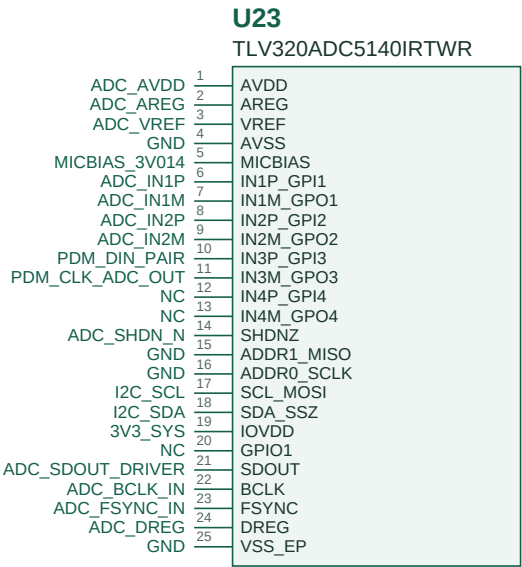
Circuit 10 / Speaker amplifier

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



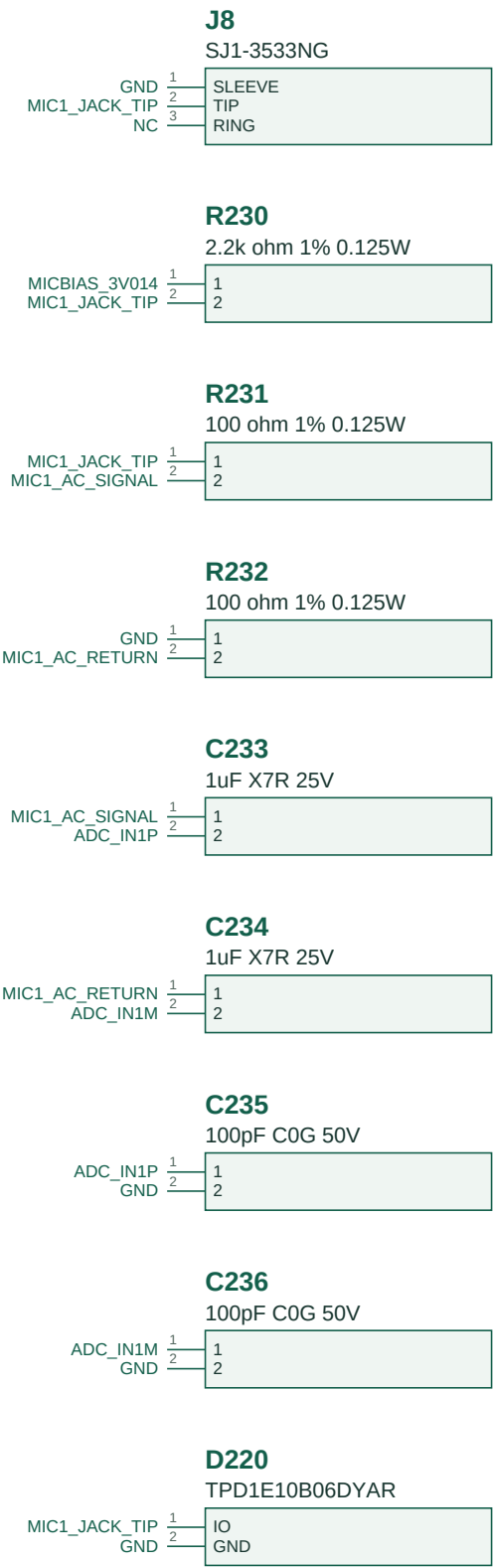
Circuit 11 / Four-channel audio capture

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



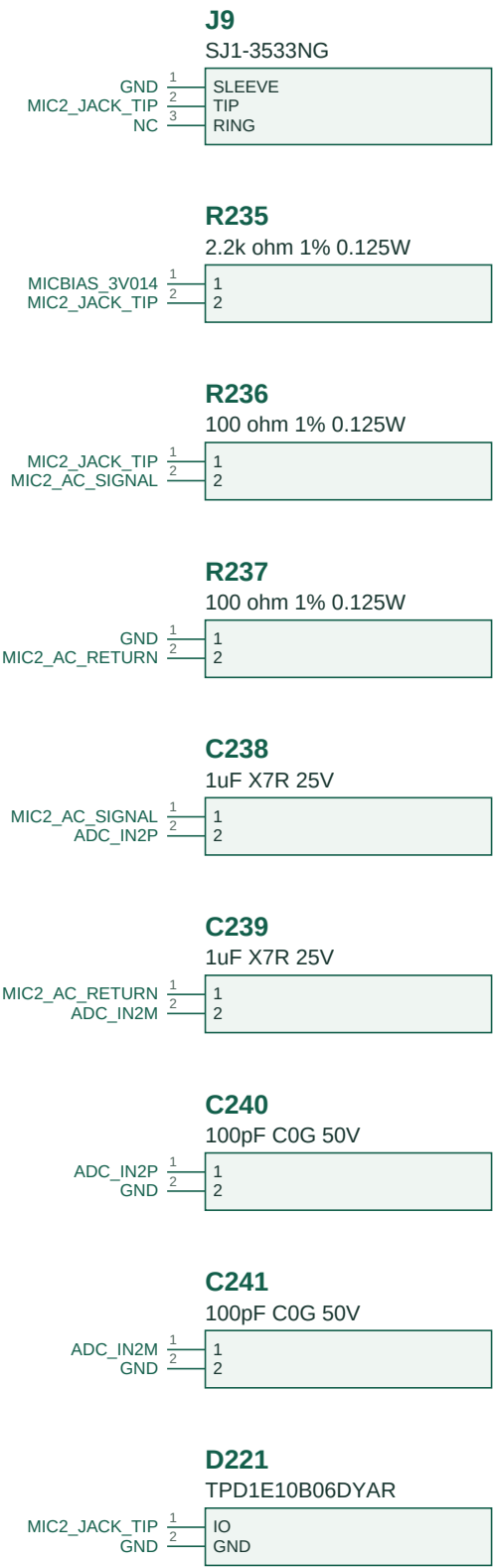
Circuit 12 / External microphone A

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



Circuit 13 / External microphone B

All physical pads shown. Global labels connect between pages. NC = intentionally unused. No production footprints assigned.



Main-board shopping list / fitted quantities

Quantities are for one device and two prototypes. DNP = zero fitted. Factory setup/attrition extras are separate; all rows still require release review.

References	One	Two	Part / value / rating	Package	Selection status
J1	1	2	GCT USB4105-GF-A	GCT USB4105 PCB-top-mount USB2 receptacle, 0.95 mm shell stakes	Selected design candidate; footprint and release review pending
F1	1	2	Littelfuse 0467002.NRHF 2 A	0603 fuse	Selected design candidate; footprint and release review pending
U2	1	2	BQ24074RGTR	VQFN16 RGT0016 + exposed pad	Selected design candidate; footprint and release review pending
R1	1	2	1.24k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
R2	1	2	2.00k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
R3	1	2	68.0k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
R4, R5, R8, R9, R10, R11, R12, R14, R18	9	18	100k	0603	HOLD - electrical specification selected; qualify exact MPN
C1	1	2	4.7uF 16V X7R	0805	HOLD - electrical specification selected; qualify exact MPN
C2, C16	2	4	10uF 10V X7R	0805	HOLD - electrical specification selected; qualify exact MPN
C3, C17, C18	3	6	22uF 10V X7R	0805	HOLD - electrical specification selected; qualify exact MPN
J2	1	2	JST S2B-PH-SM4-TB(LF)(SN) battery	JST PH 2mm 2-position right angle SMT	Selected design candidate; footprint and release review pending
J3	1	2	3-pin keyed NTC connector - JST SH family candidate	1mm SH 3pin exact drawing pending	HOLD - exact MPN and footprint pending
U3	1	2	TUSB320LAIRWBR	X2QFN12 RWB 1.6x1.6mm	Selected design candidate; footprint and release review pending
R6	1	2	887k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
R7, R13, R20, R21, R22, R25	6	12	10k	0603	HOLD - electrical specification selected; qualify exact MPN
C4, C7, C8, C9, C10, C11, C12, C19	8	16	100nF 10V	0603	HOLD - electrical specification selected; qualify exact MPN
U4	1	2	TLV75533PDBVR	SOT23-5 DBV	Selected design candidate; footprint and release review pending
C5, C6, C14, C20	4	8	1uF 10V X7R	0603	HOLD - electrical specification selected; qualify exact MPN
U5	1	2	SN74LVC1G04DBVR	SOT23-5 DBV	Selected design candidate; footprint and release review pending
U6	1	2	SN74LVC1G08DBVR	SOT23-5 DBV	Selected design candidate; footprint and release review pending
U7, U8	2	4	SN74LVC1G32DBVR	SOT23-5 DBV	Selected design candidate; footprint and release review pending

Main-board shopping list / fitted quantities

Quantities are for one device and two prototypes. DNP = zero fitted. Factory setup/attrition extras are separate; all rows still require release review.

References	One	Two	Part / value / rating	Package	Selection status
U9	1	2	LTC2951ITS8-1#TRPBF	TSOT23-8 TS8	Selected design candidate; footprint and release review pending
SW1	1	2	POWER momentary normally-open	physical switch footprint pending	HOLD - exact MPN and footprint pending
C13	1	2	1uF 10V lowleakage X7R	0805	HOLD - electrical specification selected; qualify exact MPN
U10	1	2	TPS22918DBVR	SOT23-6 DBV	Selected design candidate; footprint and release review pending
C15	1	2	10nF 25V X7R	0603	HOLD - electrical specification selected; qualify exact MPN
R15	1	2	100ohm	0603	HOLD - electrical specification selected; qualify exact MPN
U11	1	2	TPS63802DLAR	VSON-HR/HotRod DLA0010A 10-pin 2x3mm	Selected design candidate; footprint and release review pending
L1	1	2	0.47uH Coilcraft XFL4015-471ME	Coilcraft XFL4015	Selected design candidate; footprint and release review pending
R16	1	2	511k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
R17	1	2	91.0k 1%	0603	HOLD - electrical specification selected; qualify exact MPN
U12	1	2	BQ27441DRZR-G1A	VSON12 DRZ0012A 2.5x4 mm + exposed pad	Selected design candidate; footprint and release review pending
R19	1	2	0.010ohm 1% lowTCR 0.25W	0805 sense resistor	HOLD - electrical specification selected; qualify exact MPN
R23, R24, R26	3	6	1M	0603	HOLD - electrical specification selected; qualify exact MPN
C21	1	2	470nF 6.3V X7R	0603	HOLD - electrical specification selected; qualify exact MPN
U13	1	2	TPD2EUSB30DRTR	SOT-9X3 DRT 3-pin 1x0.8mm body	Selected design candidate; footprint and release review pending
D1	1	2	TPD1E10B06DYAR	DYA SOD-523	Selected design candidate; footprint and release review pending
U1	1	2	ESP32-S3-WROOM-1-N16R8	Espressif WROOM-1 18.0x25.5mm module, 40 castellated pads plus GND pad 41	Selected design candidate; footprint and release review pending
U30	1	2	RV-3028-C7-TA-QC	Micro Crystal C7 3.2x1.5x0.8mm	Selected design candidate; footprint and release review pending
U31	1	2	PCA9536DR	TI D SOIC-8 4.9x3.91mm, 1.27mm pitch	Selected design candidate; footprint and release review pending
U34	1	2	TS3USB221ARSER	TI RSE UQFN-10 2.0x1.5mm	Selected design candidate; footprint and release review pending
U35	1	2	TPD2EUSB30DRTR	TI DRT SOT-3 1.0x0.8mm, 0.50mm maximum height	Selected design candidate; footprint and release review pending
J6	1	2	DM3AT-SF-PEJM5	Hirose DM3AT push-push microSD top mount	Selected design candidate; footprint and release review pending

Main-board shopping list / fitted quantities

Quantities are for one device and two prototypes. DNP = zero fitted. Factory setup/attrition extras are separate; all rows still require release review.

References	One	Two	Part / value / rating	Package	Selection status
J7	1	2	B8B-PH-K-S(LF)(SN)	JST PH 8-position 2.0mm vertical through-hole	Selected design candidate; footprint and release review pending
R300, R301, R304, R305, R310, R311, R312, R313, R314, R321	10	20	10k	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
C300	1	2	1uF	0603 metric1608 X7R (except marked C0G), 10V minimum	HOLD - electrical specification selected; qualify exact MPN
C301, C305, C310	3	6	10uF	0603 metric1608 X7R (except marked C0G), 10V minimum	HOLD - electrical specification selected; qualify exact MPN
C302, C306, C311, C320, C321, C330, C340	7	14	100nF	0603 metric1608 X7R (except marked C0G), 10V minimum	HOLD - electrical specification selected; qualify exact MPN
R302, R303, R315	3	6	22ohm	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
C303, C304	0	0	DNP / C0G tuning footprint	0603 metric1608 X7R (except marked C0G), 10V minimum	DNP - decision pending; do not buy for normal assembly
R306, R307	2	4	33ohm	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
R316	1	2	100k	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
C312, C331, C332, C333, C334	5	10	10nF	0603 metric1608 X7R (except marked C0G), 10V minimum	HOLD - electrical specification selected; qualify exact MPN
R320	1	2	330ohm	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
R330, R331, R332, R333	4	8	47k	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
R341, R342	2	4	49.9ohm	0603 metric1608, 1%, 0.1W	HOLD - electrical specification selected; qualify exact MPN
SW300, SW301, SW302, SW303, SW304, SW305	6	12	B3U-1000P	Omron B3U 3.0x2.5mm, top-actuated 2-terminal SMD, no ground terminal or boss	Selected design candidate; footprint and release review pending
U20, U21	2	4	IM69D128SV01XTMA1	Infineon PG-TLGA-5-2, 3.50x2.65mm, bottom acoustic port	Selected design candidate; footprint and release review pending
R201, R202	2	4	100 ohm 1%	0805	HOLD - electrical specification selected; qualify exact MPN
R203, R207, R210, R211	4	8	33 ohm 1%	0805	HOLD - electrical specification selected; qualify exact MPN
R204	1	2	0 ohm	0805	HOLD - electrical specification selected; qualify exact MPN

Main-board shopping list / fitted quantities

Quantities are for one device and two prototypes. DNP = zero fitted. Factory setup/attrition extras are separate; all rows still require release review.

References	One	Two	Part / value / rating	Package	Selection status
C200, C202	2	4	1uF X7R 10V	0805	HOLD - electrical specification selected; qualify exact MPN
C201, C203, C205	3	6	100nF X7R 10V	0805	HOLD - electrical specification selected; qualify exact MPN
C204	1	2	4.7uF X7R 10V	0805	HOLD - electrical specification selected; qualify exact MPN
U22	1	2	MAX98357AETE+	TQFN16 3x3mm exposed-pad T1633+4; 21-0136/90-0031	Selected design candidate; footprint and release review pending
R208, R209	2	4	100k 1%	0805	HOLD - electrical specification selected; qualify exact MPN
C206	1	2	10uF X7R 10V	0805	HOLD - electrical specification selected; qualify exact MPN
C207	0	0	47uF X7R 10V DNP	1210	DNP - decision pending; do not buy for normal assembly
J5	1	2	SM02B-SRSS-TB	JST SH 1.0mm 1x2 right-angle SMD	Selected design candidate; footprint and release review pending
U23	1	2	TLV320ADC5140IRTWR	TI RTW0024A WQFN24 4x4mm 0.5mm pitch, exposed ground pad 25	Selected design candidate; footprint and release review pending
U24	1	2	SN74LVC1G17DBVR	TI DBV0005A SOT-23-5	Selected design candidate; footprint and release review pending
FB220	1	2	BLM21PG221SN1D	0805	Selected design candidate; footprint and release review pending
R220, R224	2	4	100k ohm 1% 0.125W	0805	HOLD - electrical specification selected; qualify exact MPN
R221, R222, R223	3	6	33 ohm 1% 0.125W	0805	HOLD - electrical specification selected; qualify exact MPN
C220, C233, C234, C238, C239	5	10	1uF X7R 25V	0805	HOLD - electrical specification selected; qualify exact MPN
C221, C223, C225, C227, C229, C230, C232	7	14	100nF X7R 25V	0805	HOLD - electrical specification selected; qualify exact MPN
C222, C224, C231	3	6	10uF X7R 16V	0805	HOLD - electrical specification selected; qualify exact MPN
C226, C228	2	4	2.2uF X7R 16V	0805	HOLD - electrical specification selected; qualify exact MPN
J8, J9	2	4	SJ1-3533NG	Same Sky SJ1-3533NG, right-angle 3-pin through-hole TRS	Selected design candidate; footprint and release review pending
R230, R235	2	4	2.2k ohm 1% 0.125W	0805	HOLD - electrical specification selected; qualify exact MPN
R231, R232, R236, R237	4	8	100 ohm 1% 0.125W	0805	HOLD - electrical specification selected; qualify exact MPN
C235, C236, C240, C241	4	8	100pF C0G 50V	0805	HOLD - electrical specification selected; qualify exact MPN

Main-board shopping list / fitted quantities

Quantities are for one device and two prototypes. DNP = zero fitted. Factory setup/attrition extras are separate; all rows still require release review.

References	One	Two	Part / value / rating	Package	Selection status
D220, D221	2	4	TPD1E10B06DYAR	TI DYA0002A SOD-523	Selected design candidate; footprint and release review pending

Primary references and release scope

Manufacturer datasheets control selected pins and limits. Source notes retain the calculations and assumptions.

01 https://documentation.espressif.com/esp32-s3-wroom-1_wroom-1u_datasheet_en.pdf

02 https://files.waveshare.com/upload/0/0c/1.3inch_LCD_Module_Schematic.pdf

03 <https://gct.co/files/drawings/usb4105.pdf>

04 https://omronfs.omron.com/en_US/ecb/products/pdf/en-b3u.pdf

05 <https://www.analog.com/media/en/technical-documentation/data-sheets/295112fb.pdf>

06 <https://www.analog.com/media/en/technical-documentation/data-sheets/max98357a-max98357b.pdf>

07 <https://www.hirose.com/product/download/?distributor=chip1&lang=en&num=DM3AT-SF-PEJM5&type=2d>

08 <https://www.infineon.com/assets/row/public/documents/24/49/infineon-im69d128s-datasheet-en.pdf>

09 <https://www.jst-mfg.com/product/pdf/eng/ePH.pdf>

10 <https://www.jst.com/products/crimp-style-connectors-wire-to-board-type/sh-connector>

11 <https://www.littelfuse.com/assetdocs/fuse-467-datasheet?assetguid=4a59f034-1cca-460e-a5ba-e1e66247c76d>

12 https://www.microcrystal.com/fileadmin/Media/Products/RTC/App.Manual/RV-3028-C7_App-Manual.pdf

13 <https://www.murata.com/en-eu/api/pdfdownloadapi?cate=cgsubChipFerriBead&partno=BLM21PG221SN1D>

14 <https://www.sameskydevices.com/product/resource/sj1-353xng.pdf>

15 <https://www.ti.com/document-viewer/SN74LVC1G04/datasheet/GUID-047EF317-EC05-483E-B17F-7CDF4F4C03DB>

16 <https://www.ti.com/lit/ds/symlink/bq24074.pdf>

17 <https://www.ti.com/lit/ds/symlink/bq27441-g1.pdf>

18 <https://www.ti.com/lit/ds/symlink/pca9536.pdf>

Primary references and release scope

Manufacturer datasheets control selected pins and limits. Source notes retain the calculations and assumptions.

19 <https://www.ti.com/lit/ds/symlink/sn74lvc1g08.pdf>

20 <https://www.ti.com/lit/ds/symlink/sn74lvc1g32.pdf>

21 <https://www.ti.com/lit/ds/symlink/tlv320adc5140.pdf>

22 <https://www.ti.com/lit/ds/symlink/tlv755p.pdf>

23 <https://www.ti.com/lit/ds/symlink/tpd1e10b06.pdf>

24 <https://www.ti.com/lit/ds/symlink/tpd2eusb30.pdf>

25 <https://www.ti.com/lit/ds/symlink/tps22918.pdf>

26 <https://www.ti.com/lit/ds/symlink/tps63802.pdf>

27 <https://www.ti.com/lit/ds/symlink/ts3usb221a.pdf>

28 <https://www.ti.com/lit/ds/symlink/tusb320lai.pdf>

29 <https://www.ti.com/lit/gpn/sn74lvc1g17>

30 <https://puiaudio.com/file/specs-AOM-5024L-HD-R.pdf>

31 <https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/i2s.html>

32 <https://www.pcbway.com/pcb-assembly.html>

33 <https://www.silvtronics.com/>

34 https://files.waveshare.com/upload/0/0c/1.3inch_LCD_Module_Schematic.pdf